

Spring AI: A Game Changer in Java Programming

Spring AI is transforming Java programming by enabling developers to build AI-infused solutions seamlessly.

Spring AI provides a comprehensive framework that simplifies the integration of AI capabilities into Java applications. By leveraging its power, developers can create intelligent solutions with minimal overhead, ensuring efficiency and scalability.

Chat completion models and SpringAI

Chat completion models, powered by natural language processing (NLP), enhance user interactions by predicting and generating human-like responses. In a Java Spring Boot application, you can utilise Spring AI to integrate a chat completion model to augment customer service. For instance, in the banking sector, a chatbot can be developed to assist customers with their queries, providing timely and accurate responses based on historical data and context.

As an example, the following code will create a service that uses the ChatAI component to generate a response based on the user's input message.

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import com.springai.chat.CompletionRequest;
import com.springai.chat.CompletionResponse;
import com.springai.chat.ChatAI;
@Service
public class ChatCompletionService {
    @Autowired
    private ChatAI chatAI;
```



```
public String getChatCompletion(String
userMessage) {
    CompletionRequest request = new
CompletionRequest(userMessage);
    CompletionResponse response =
chatAI.complete(request);
    return response.getCompletion();
}
}
```

Embedding models

Embedding models transform text into numerical vectors that capture the semantic meaning of words. These embeddings can be employed for various tasks such as document similarity, recommendation systems, and sentiment analysis. Within a retail application, Spring AI can be used to implement a recommendation engine that suggests products to customers based on their past purchases and

browsing behaviour.

The example code given below creates a Spring service that processes the sentiment of the given document text using the SentimentAI component of the Spring AI framework.

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import com.springai.sentiment.SentimentRequest;
import com.springai.sentiment.SentimentResponse;
import com.springai.sentiment.SentimentAI;
@Service
public class SentimentAnalysisService {
    @Autowired
    private SentimentAI sentimentAI;
    public String analyzeSentiment(String
documentText) {
        SentimentRequest request = new
SentimentRequest(documentText);
        SentimentResponse response =
sentimentAI.analyze(request);
        return response.getSentiment();
    }
}
```

Text to image model

The text to image model generates images from textual descriptions, opening new avenues for creative and engaging content. In the insurance industry, this can be particularly useful for generating visual representations of complex policy documents. Spring AI enables developers to incorporate this functionality, allowing users to visualise policy details effortlessly.